

IoT Weather Station

A Three-Tier Cloud-Connected Monitoring System

Halim YOUSSEF

Student ID: 18818153

Cloud & Advanced Analytics · May 2026



Motivation



Poor air quality is invisible

- TVOC and eCO₂ are odourless — they accumulate silently in poorly ventilated rooms
- No perceptible signal until symptoms appear (fatigue, headaches, poor concentration)
- ENV3 sensor tracks both gases continuously, readings pushed every 5 minutes
- Colour-coded quality badges give an instant status at a glance
- Delta indicator vs 1 hour ago shows whether air is improving or worsening



We check the weather when it's already too late

- Most people only look up the forecast when they're about to leave — too late to adapt
- Outdoor conditions pulled from OpenWeather every 5 minutes: temp, humidity, wind, UV, precipitation
- Always visible on both the dashboard and the M5Stack device without opening any app
- Ambient display means the decision to grab an umbrella happens naturally, not reactively



We miss trains because we checked too late

- Train info exists — but it lives in a separate app you only open when already rushing
- Switching context mid-morning-routine adds friction at the worst possible moment
- Live SBB departures integrated directly alongside weather and air quality data
- Countdown timers show minutes remaining to each departure in real time
- Commute awareness becomes passive — absorbed on your way out, not actively checked

Solution at a Glance

M5Stack device

ENV III, Air Quality, PIR motion sensors

4-screen touch UI (Home / Forecast / Trains / Actions)

Motion-triggered Gemini-phrased announcements

Flask backend · Cloud Run

All external API calls & BigQuery writes

SHA-256 device authentication

In-memory weather + BigQuery cache

Streamlit dashboard · Cloud Run

5 pages: Overview · Forecast · Trains · AI · History

Animated weather overlays

Session-affinity Cloud Run deployment

Gemini AI voice assistant

Natural-language sensor & forecast queries

BigQuery + Forecast + Train tool-use

Google STT input · Google TTS output

Tech Stack & Services

HARDWARE

M5Stack Core2
main device

ENV III sensor
temp & humidity

Air Quality sensor
TVOC / eCO₂

PIR motion sensor
announcement trigger

GOOGLE CLOUD

Cloud Run
backend service

Cloud Run
dashboard service

BigQuery
data warehouse

Vertex AI
Gemini 2.5 Flash

STT / TTS
voice assistant

EXTERNAL APIS

OpenWeather API
outdoor conditions

transport.opendata.ch
train departures

APP STACK

Python 3.12 / Flask
backend API

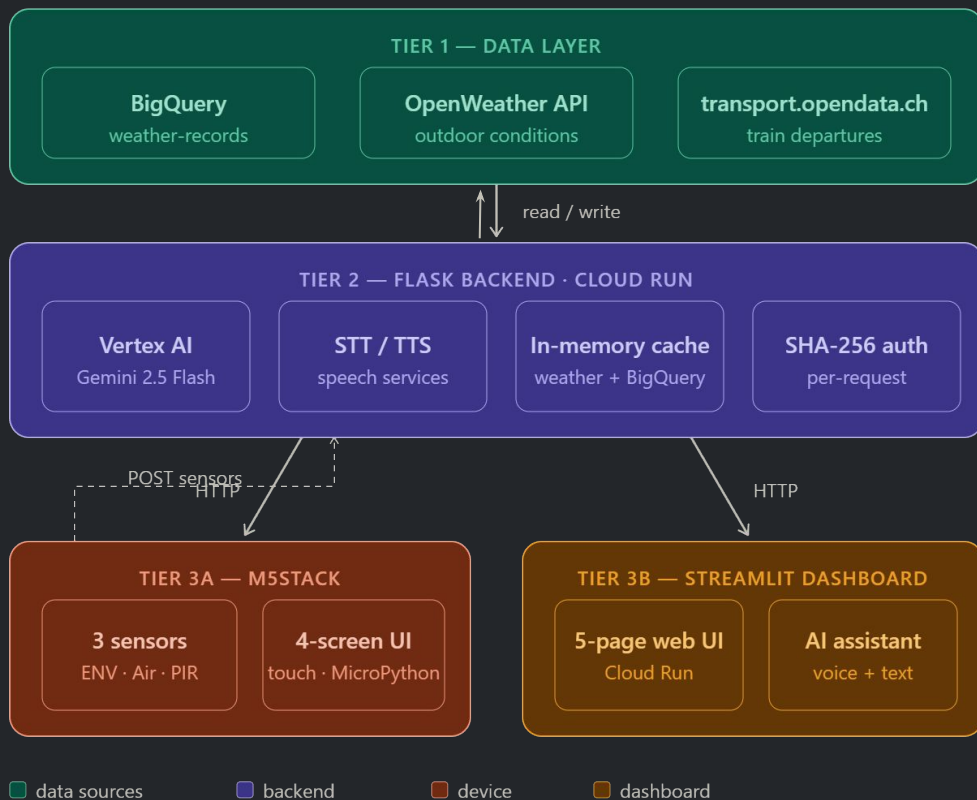
Streamlit
dashboard UI

MicroPython / UIFlow
M5Stack firmware

BigQuery client
data layer access

Docker / Cloud Build
containerised deploy

Architecture — Three-Tier Overview



PORTA

PORTC

PORTB

Actions

M5Stack Core2 — Demo



M5Stack Core2 — Hardware

ISSUES ENCOUNTERED

Speaker Volume Very Low

Works correctly in the dashboard... harder issue ?

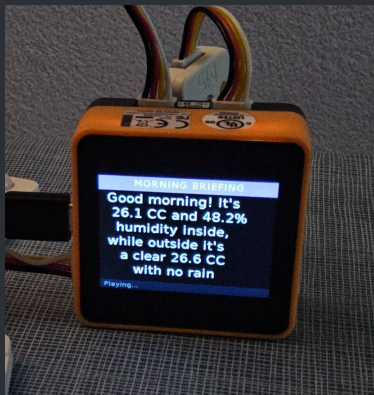


Microphone input not captured

always error `"silent:mic_unav"` or `"signal: 0 SILENT !"`



Work Around → Actions Screen



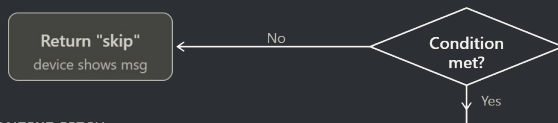
TRIGGER



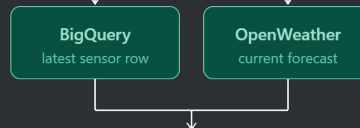
REQUEST



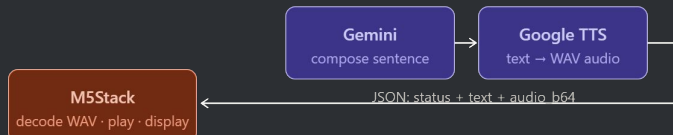
CONDITION GATE



CONTEXT FETCH



SYNTHESIS



Weather Dashboard

Multiple Pages

Overview

Live Indoor and Outdoor metrics with dynamic animation

Anx A

Forecast

5-days and 18h predictions

Anx B

Trains

Real-time SBB train departure board with countdown timers

Anx C

AI Assistant

AI Q&A w/ TTS and SST + Quick Action buttons

Anx D

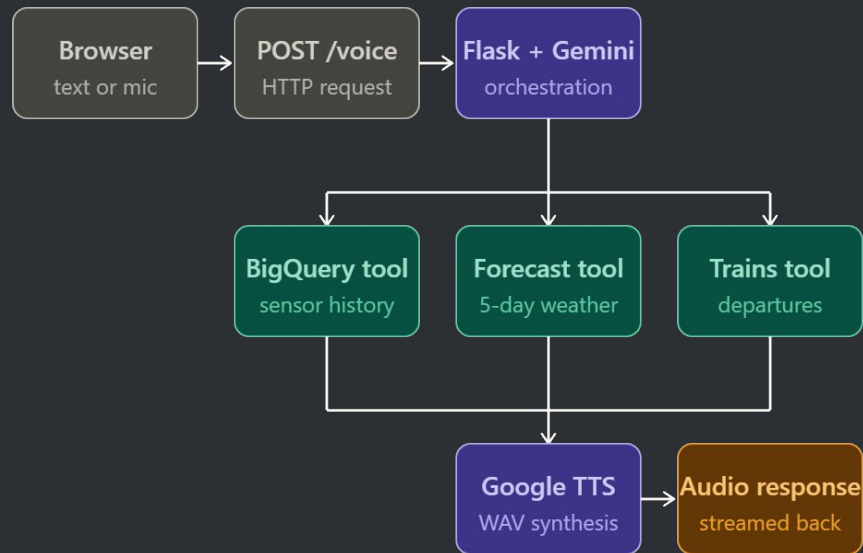
History

Time-series plot with date-range

Anx E

► Dashboard demo

VOICE QUERY FLOW



Next Steps

1

Hardware fixes

Identify correct I2C targets to fix the speaker volume and enable the microphone.

2

Full on-device voice loop

Bring the complete STT → Gemini → TTS pipeline directly onto the M5Stack.

3

Mobile Push notifications

Integrate for real-time air-quality threshold alerts beyond the dashboard

4

Extended forecast

Upgrade to 7-day forecast via OneCall API 3.0.

5

3D-printed enclosure

Design and print a custom housing for the device and all attached sensors.

Thank You !

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GitHub github.com/halim-y/cloud-project

Dashboard [cloud-project-dashboard](#)

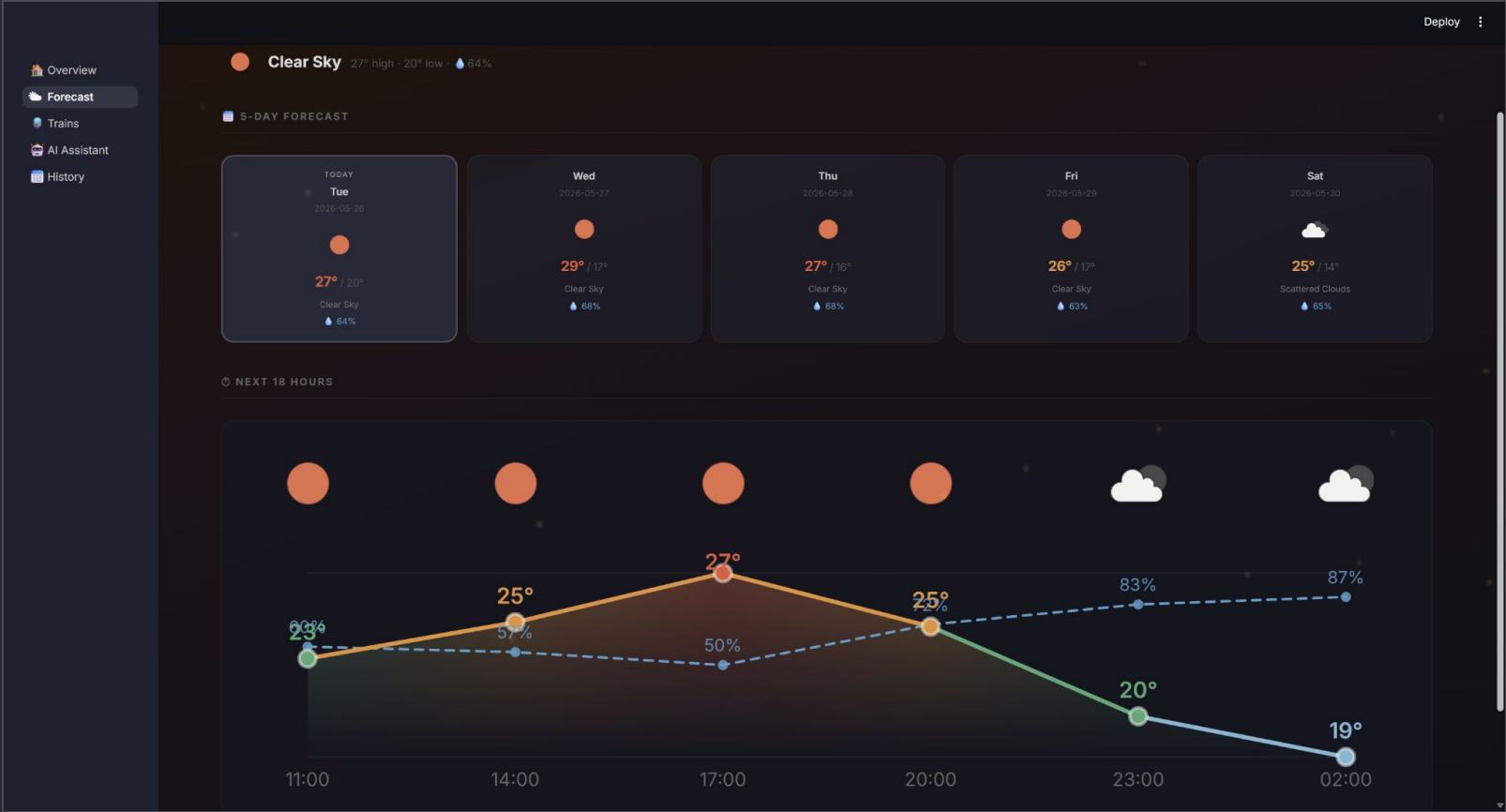
ANNEXES

Additional detail slides

Annex A — Overview



Annex B — Forecast



Annex C — Trains

Overview

Forecast

Trains

AI Assistant

History

Next Departures

ROUTE

From

Genève

To

Renens VD

Type

All

Refresh

GENÈVE → RENENS VD

Updated at 10:17 · auto-refreshes every 60 s · click a card to expand

GENÈVE

NEXT DEPARTURE

10:24 → 11:03 39 min IR 95 Pl. 4

10:17

in 7 min

39 min → Direct

10:24 Genève

11:03 Renens VD

RENENS VD

7 MIN

IR 95 → Brig

10:35 → 11:07 32 min IC 1 Pl. 5

On time in 18 min

10:38 → 11:23 45 min RE 33 Pl. 3

On time in 21 min

Annex D — AI Assistant

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Overview

Forecast

Trains

AI Assistant

History

Deploy

AI Assistant

Ask about your home environment in plain language — type a question, use a quick action, or speak.

Quick actions

ASK

What's the temperature now?

How is the air quality this week?

Is it raining today?

What's the humidity trend?

ANNOUNCE — proactive announcements the MSStack plays on motion

Auto-pick

Current summary

Morning briefing

Rain reminder

Humidity alert

Train alert

CONVERSATION

Do I need an umbrella this week ?

No, the forecast for this week shows clear skies and scattered clouds, so you won't need an umbrella.

Listen

MSSTACK - HUMIDITY ALERT

skipped — condition not met

Type your question...

Send

Tip: click the mic, wait for the icon to flip to , then speak. The browser needs a moment to enable the microphone.

Annex E — History

